IGCSE 0607 Practice Questions — E7.4

Trigonometric functions

Total Questions: 9 — **Total Marks:** 48 Marks

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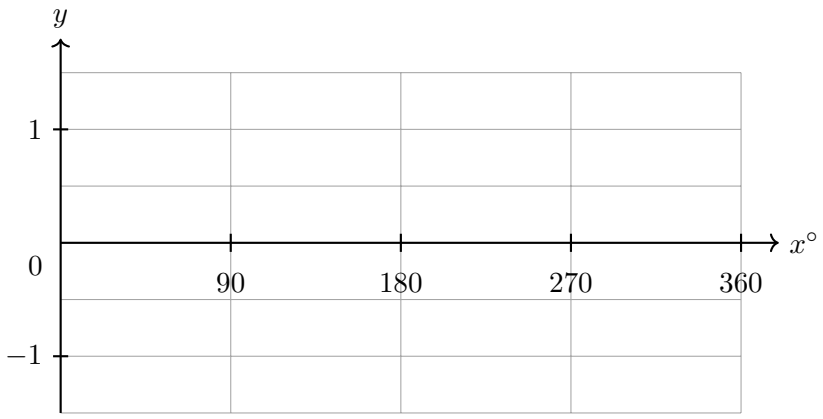
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1 Solve the equation for $0° \leq x \leq 360°$.

$\sqrt{2}\sin x - 1 = 0$

$x =$ ..... **[3]**

2 **(a)** On the grid, sketch the graph of $y = \sin x$ for $0° \leq x \leq 360°$.



[2]

(b) Write down the exact solutions to the equation $\sin x = 0.5$ for $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

3 Answer the following questions for $0° \leq x \leq 360°$.

(a) Solve the equation

$$2\cos x - 1 = 0$$

$x =$ ..... **[3]**

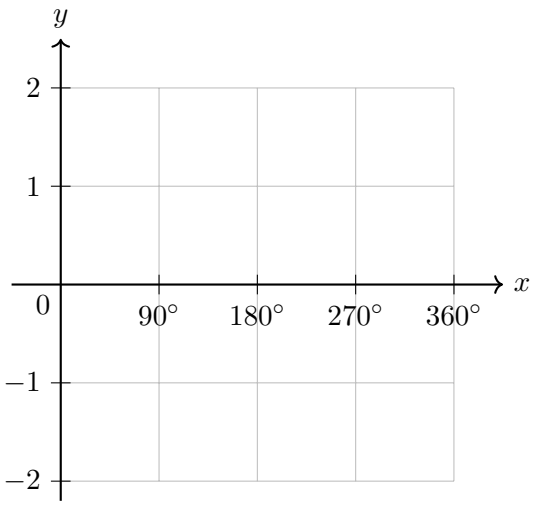
(b) Write down the number of solutions to the equation

$$3\sin x = 5$$

..... **[1]**

4 The properties of trigonometric graphs can be used to solve equations.

(a) On the grid, sketch the graph of $y = \cos x$ for $0° \leq x \leq 360°$.



[2]

(b) Write down the coordinates of the minimum point on this graph.

(..... , ...) **[1]**

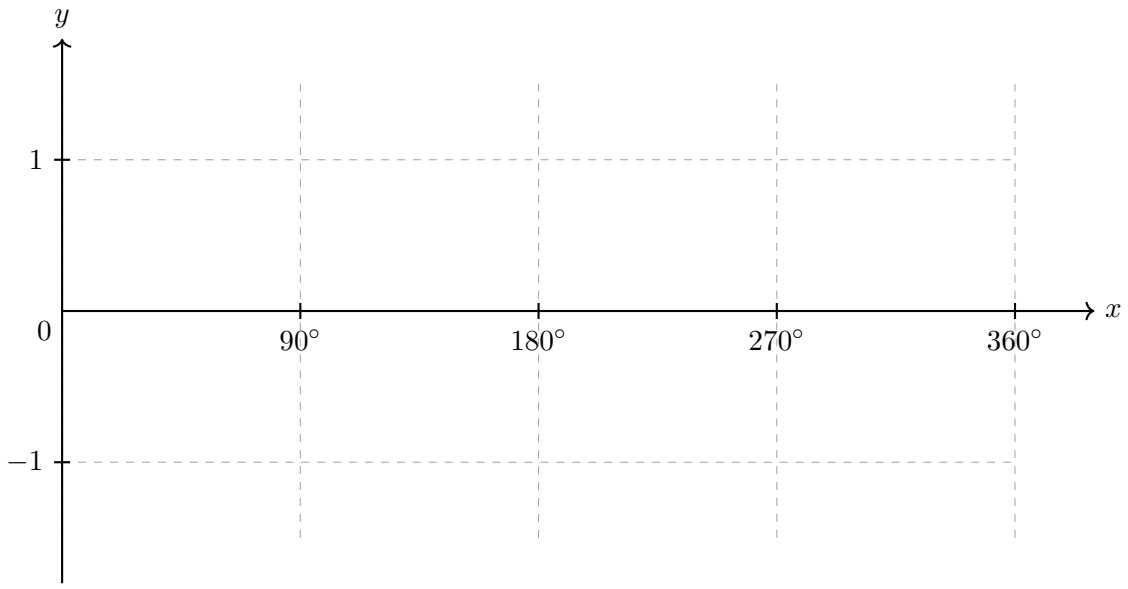
(c) Solve the equation for $0° \leq x \leq 360°$.

$2\cos x + 1 = 0$

$x =$ ..... **[3]**

5 The functions $g(x) = \sin x$ and $h(x) = \cos x$ are defined for $0° \leq x \leq 360°$.

(a) On the grid provided, sketch the graphs of $y = g(x)$ and $y = h(x)$.



[3]

(b) Solve the equation $g(x) = h(x)$ for $0° \leq x \leq 360°$.

$x =$ ..... **[2]**

(c) Use your graphs to write down the range of values of x for which $g(x) > h(x)$ for $0° \leq x \leq 360°$.

..... **[2]**

6 Consider the trigonometric function $f(x) = 4\sin x + 3$ for $0° \leq x \leq 360°$.

(a) Solve the equation $f(x) = 0$.
Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

(b) Write down the exact coordinates of the minimum point on the graph of $y = f(x)$.

(..... , ...) **[2]**

7 **(a)** Solve the equation for $0° \leq x \leq 360°$.

$4\cos x + 1 = 0$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

(b) Solve the equation for $0° \leq x \leq 360°$.

$2\tan x - 5 = 0$

Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

8 **(a)** The equation $a \sin x - b = 0$ has a solution $x = 30°$.
Given that a and b are positive integers with no common factors, find the value of a and the value of b.

$a =$ ..... , $b =$ ..... **[2]**

(b) Write down the second solution to the equation $a \sin x - b = 0$ for $0° \leq x \leq 360°$.

$x =$ ..... ° **[1]**

(c) A horizontal line $y = k$ intersects the graph of $y = \cos x$ at exactly two points in the interval $0° \leq x \leq 360°$.
The x-coordinate of one of these points is $242°$.
Find the value of k correct to 3 significant figures and find the x-coordinate of the second point of intersection.

$k =$ ..... , $x =$ ..... **[3]**

9 Algebraic techniques can be applied to solve trigonometric equations.

(a) Solve the equation for $0° \leq x \leq 360°$.

$$5\tan x + 14 = 0$$

Give your answers correct to 1 decimal place.

$x =$ ..... **[3]**

(b) Solve the equation for $0° \leq x \leq 360°$.

$$2\sin^2 x - \sin x = 0$$

$x =$ ..... **[4]**